Alexander Shaham

Los Angeles, CA • ajshaham@gmail.com • portfolio.ajshaham.workers.dev • github.com/Nuja1337

EXPERIENCE

Independent Game Developer — Unannounced multiplayer RPG Feb 2026 – Present

Classless arena-and-dungeon RPG for PC · Godot 4, C#, .NET 10, Steamworks · playable online

- Built a server-authoritative, deterministic 60 Hz simulation as a plain .NET library with no engine dependency, so the same code runs in the headless dedicated server and under xUnit. One CPU core hosts 41 concurrent matches.
- Cut each client's worst-second bandwidth from 198 to 80 kbps (~60%) with delta snapshots over Steam Datagram Relay, and added lag compensation and reconnect.
- Filtered snapshots by team vision on the server so clients never receive hidden entities, which rules out map-hack cheats.
- Implemented the combat runtime (ability state machine, projectiles, zones, interrupts, crowd control with diminishing returns), bot AI, and an in-game dev console.
- Set the combat readability rules: colorblind-safe colors, lightness before hue, and one telegraph style for players and bosses.
- Built an ASP.NET Core matchmaking service with hidden player ratings.
- Wrote 5,000+ xUnit tests, including deterministic golden replays verified by state hash and zero-allocation checks.
- Prototyped in Unreal Engine 5 (C++), then migrated to Godot 4 and C# (.NET 10) so the simulation could run headless as plain .NET, which was far harder in UE5. The move also sped up iteration, debugging, and AI-assisted development.

HCL BigFix — Software Engineer II (converted from Software Engineering Intern) May 2023 – Feb 2025

Joined as an intern and did the same engineering work throughout; full-time conversion was delayed by a hiring freeze.

- Built a React + Material UI (MUI) component library adopted by 100+ engineers across 7+ apps, including teams in Rome and Bangalore. Cut duplicated UI code by ~32%.
- Shipped 100+ versioned, documented components at 95%+ test coverage, plus a Storybook-style workbench used by engineering, QA, and design to review components.
- Reduced component re-renders by ~60% by moving global state to Zustand.
- Expanded security-platform coverage by 66% by integrating Tenable SC and IO into the ingestion pipeline (Node.js, SQL, REST).
- Delivered major features to alpha, beta, and stable releases through CI/CD (GitHub Actions, Jenkins) with <5% regression, on a plugin-based architecture that let teams ship independently.
- Reduced configuration errors by ~13% by redesigning the Insights datasource setup flow with UX and backend teams.
- Mentored 4 engineers across teams on onboarding, design-system usage, and architectural standards.

ADDITIONAL EXPERIENCE

Independent Trader — Self-employed Feb 2025 – Feb 2026

- Traded stocks full time with personal capital. Wrote Python scripts to pull price data, backtest setups, parse broker trade exports, and send alerts. Used AI tools for company and earnings research.

Freelance Web Developer — Accessible, multilingual React landing page for a healthcare clinic Sep 2022 – Jan 2023

PROJECTS

Guild Golem | TypeScript, React Router, Hono, Cloudflare Workers, D1, R2 guildgolem.com

- Guild management site for World of Warcraft: Forever: raid calendar with Discord sign-ups, raid planner, roster and attendance, three loot systems (council, EPGP, DKP), and a 13-command Discord bot.
- Includes a combat-log parser running in a Web Worker and a simulator covering 23 specs on game-client data. Covered by 1,000+ tests, with a strict CSP and bundle-size budgets checked before every deploy.

Drafts | TypeScript, React, Hono, Durable Objects, D1, KV, Workers AI drafts.ajshaham.workers.dev

- Real-time League of Legends and Dota 2 draft rooms that suggest picks and bans from matchups and synergies, with an AI coach.

Tally | JavaScript, IndexedDB, PWA, Cloudflare Workers, D1 tally.ajshaham.workers.dev

- Framework-free, local-first training and nutrition PWA. Works offline and syncs across devices, merging conflicting edits.

Shipped Games | Keepie Uppies (Google Play), Queen of the Hill (itch.io)

- Released 2 small games: a Flappy Bird-style mobile game and a Cookie Clicker-style idle game.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript, C#, C++, Python, SQL, Java, GDScript
Frontend	React, React Router, Next.js, Remix, Zustand, Material UI, Tailwind, HTML, CSS, SPA, PWA, Web Workers, IndexedDB
Backend	Node.js, Hono, Express, ASP.NET Core (.NET 10), REST APIs, WebSockets, PostgreSQL, SQLite, pandas
Cloud & DevOps	Cloudflare Workers, Pages, D1, R2, KV, Durable Objects, Workers AI, Wrangler, Docker, Linux
Tooling & CI	Vite, Vitest, Jest, xUnit, Playwright, GitHub Actions, Jenkins, Git
Game Dev	Godot 4 (.NET), Unreal Engine 5 (C++, Blueprints, GAS), Unity (C#), Phaser 3, Steamworks, Steam Datagram Relay
AI & Tools	AI-assisted development (Claude Code, MCP), Figma, Jira, YouTrack, Android SDK, Discord API (discord.js)
Spoken	English (native), Spanish (proficient)

EDUCATION

University of California, Santa Cruz — B.S. Computer Science: Computer Game Design Sep 2017 – Jun 2022